

Logos and Theming

From an image's identity to useful application themes

A detailed project overview

Architecture, color decisions, application contracts and recoverable consolidation.



One pipeline, three kinds of decisions

A color can be faithful to an image and still be wrong for a role.

Extraction

Which observed colors represent the image?

Meaning

Which colors can serve as surfaces, text, accents and status signals?

Application behavior

Which combinations remain useful in each target's native constraints?

Consolidation preserves the history

The new projects organize work that grew in several places.

Former location	Disposition
Lang-Theme-Generator	Engine and legacy generation history
Aptlantis Logos	Artwork, catalogs and converters
.cts_holding/palette-transformer	Transformation tools and differing versions
Documents working collection	Historical outputs, source assets and configuration

Ownership is part of the design

A shared workflow does not require one project to own everything.

Owner	Responsibility
Logos-And-Theming	Theme engine, artwork/catalogs, ICO/SVG converters, atlas
SESM-Metadata-Embedder	SVG metadata, configuration, schema checks, embedding tests
City Hall	Canonical standards and schemas
Public asset copies	Existing publication and deployment state

The six-stage flow

Every boundary produces something inspectable.

Stage	Decision / output
1 · Image	Preserved bytes, hash, normalized pixels
2 · Candidates	Up to 192 observed representatives and population weights
3 · Canonical 32	Stable IDs, four palette formats, selection diagnostics
4 · Semantic roles	Assignments, hue suitability and collision findings
5 · Target adaptation	Derived tokens, aliases, origins and omissions
6 · Themes + review	Native files, validation and self-contained HTML

Use a specialist tool at each stage

The pipeline combines strengths instead of asking one tool to infer everything.

Tool	Job
ImageMagick	Decoding, orientation, ICC-aware sRGB normalization
NumPy + scikit-learn	Sampling and population-weighted OKLab clustering
ColorAide	Color conversion, contrast and sRGB gamut handling
Python + PowerShell	Pipeline contracts, configuration and local orchestration
Artifact Tool / Open XML	Editable PowerPoint content and package checks

Normalize pixels; preserve the original

Normalization creates a working representation without replacing the source.

Interpret the image

Apply orientation and ICC-aware sRGB conversion through ImageMagick.

Respect transparency

Exclude fully transparent pixels; weight partial opacity by alpha.

Bound the computation

Default sampling limit: 65,536 pixels. Fixed seed and stable ordering.

Candidates are observed colors

A cluster center guides selection; it does not become an invented export color.

Perceptual coordinates

Convert sampled sRGB colors to OKLab for clustering.

Population matters

Weight clusters by image population and partial opacity.

Representative selection

Choose an observed color nearest each cluster center, then order stably.

The Black-Gold canonical 32

Perceptual clustering, light/dark anchors and diversity; IDs carry no meanings.

color_01 #000000	color_02 #01060E	color_03 #051019	color_04 #0A1924
color_05 #21211C	color_06 #022742	color_07 #2F2921	color_08 #413422
color_09 #0B3F57	color_10 #664024	color_11 #18556A	color_12 #5B5752
color_13 #845328	color_14 #35737D	color_15 #0279A3	color_16 #817763
color_17 #A66B35	color_18 #4A9FAE	color_19 #A98C76	color_20 #C7833D
color_21 #89B2A7	color_22 #DC9848	color_23 #24BDD8	color_24 #D6AE7E
color_25 #EDAF57	color_26 #C0C8A9	color_27 #73DDEA	color_28 #F8C265
color_29 #FBD484	color_30 #AAF1F3	color_31 #FDEFB4	color_32 #FCFDF7

Semantic mapping makes intent explicit

Assignments can be reviewed and overridden independently of extraction.

Role	Source ID	Canonical color
background	color_04	#0A1924
foreground	color_32	#FCFDF7
primary	color_28	#F8C265
secondary	color_06	#022742
warning	color_29	#FBD484
cyan	color_27	#73DDEA

Image identity can conflict with conventions

Do not create an unrelated hue just to fill a familiar label.

A warm warning is available

Black-Gold maps warning to color_29 · #FBD484.

Magenta is unavailable

The nearest image hue is used and the mismatch is reported.

Aliases are information

Different named roles may share a color; collisions must stay visible.

Derivation changes use, not origin

Canonical colors remain untouched when applications need readable variations.

Start from a source ID

Record the role, source color and requested perceptual values.

Adjust for an actual need

Change lightness or reduce chroma for text, surfaces and states.

Keep hue intent

Fit sRGB by reducing chroma at fixed lightness and hue; record the result.

The background regression taught a boundary

The darkest extracted color is not automatically the best dark surface.

What looked wrong

Repeated near-black anchors pulled themes toward flat black backgrounds.

What changed

Coalesce the noise floor; map an identity surface near the configured lightness.

The Black-Gold result

Background color_04 · #0A1924. Pure black stays in the canonical palette.

Four targets, four native contracts

The adapter serves real application settings rather than a generic color count.

Target	Budget	Native output
Windows Terminal	20	Scheme JSON: 16 ANSI + 4 application roles
SiYuan	64	Dark CSS, metadata, README, previews, ZIP
Typora	40	CSS for document, code, sidebar and states
PowerPoint	24	12-slot color XML + editable sample deck

Names and unique colors are different counts

Black-Gold uses only the variations its current adapters need.

Target	Budget	Named	Unique
Windows Terminal	20	20	17
SiYuan	64	62	40
Typora	40	40	36
PowerPoint	24	22	18

Terminal reduction protects identity

A small vocabulary must support both meaning and readable text.

Protect required roles

Background, foreground, cursor, selection and the 16 ANSI slots.

Carry the image into text

Retain visible gold and cool cyan/blue relationships through adaptation.

Expose compromise

Report missing conventional hues, reused colors and actual contrast pairs.

Editors need content and state coverage

SiYuan and Typora require more than a page background and link color.

Area	What must hold together
Document	Body text, headings, links, quotes, tables
Code	Inline code, blocks and syntax roles
Navigation	Sidebar, muted text and active items
Interaction	Selection, focus, hover and common states
Delivery	Target CSS scopes; SiYuan package assets and metadata

PowerPoint: color theme and editable proof

Twelve native slots become a reusable visual vocabulary.

The pipeline sample

Four editable slides: title, content, Office slots and a native chart.

This overview

The same Black-Gold slots, Arial and canvas applied to project content.

Compatibility lesson

XML can parse while QName-valued metadata still breaks PowerPoint.

The run leaves a reviewable trail

A complete output directory explains where its colors came from.

Artifact	Purpose
source/ + provenance.json	Original image bytes, hash and decoding provenance
Candidates / selection	Representatives and selection diagnostics
palette.toml / txt / css / png	Exactly 32 canonical colors in reusable forms
semantics.json / tokens.json	Roles, derivations, aliases and findings
validation.json / review.html	Checks and a self-contained visual review

The command contract keeps runs bounded

Generate and import are local, reviewable operations.

Generate

`apt-theme generate IMAGE --output DIRECTORY`

Import and configure

Import exactly 32 colors; choose targets, TOML settings and semantic overrides.

Complete with findings

0: completed · 1: processing failure · 2: strict contrast failure / invalid syntax

Related utilities stay independently callable

Theme generation, conversion and metadata embedding have different jobs.

Utility	Preserved behavior
ICO converter	ImageMagick multi-frame icon conversion
SVG converter	Embedded raster by default; explicit VTracer tracing
Atlas	Artwork/catalog browsing and palette utility
SESM embedder	Preview by default; explicit write / replace and schema checks

Recovery is a deliverable

Consolidation was gated on recoverability, not just copying files.

Inventory and archive

Hash source, assets, historical outputs, configuration and Git history.

Verify destinations

Check migrated tools, artwork bytes and restored archive content.

Retire only after gates

Update references; check resolved paths and reject changed source inventories.

What the evidence establishes

Structural success, native compatibility and aesthetic quality are separate gates.

Evidence	Limit
20 focused tests after repair	Dated engineering evidence, not a new run today
Black-Gold contrast checks	Only the combinations implemented in validation
PowerPoint no-repair opening	Opening/export differs from manual edit acceptance
Coverage and visual report	Diagnostics do not prove aesthetic quality
Other native / future targets	Live acceptance remains separately recorded

A document suite, written in useful order

Keep this deck as the entry point; put depth in focused, source-backed volumes.

First · operate and explain

Runbook, artifact reference and the two regression case studies.

Next · design and extend

Extraction methods, semantic mapping, adapter contracts and native acceptance.

Then · preserve and integrate

Asset lineage, SESM companion, recovery and the living handbook.